Using Random Network Distillation Prediction Error as a Convergence Indicator in Reinforcement Learning

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https://github.com/JannisKst/rnd-convergence-rl

1 Motivation

Training reinforcement learning agents is computationally expensive, and it is often difficult to determine when training has effectively converged. Most implementations simply train for a fixed number of environment interactions, which may waste computation or stop training too early. In this project, we investigate whether the prediction error of Random Network Distillation (RND), originally introduced as an exploration bonus, can instead serve as a reliable indicator of learning progress and convergence. If successful, such a signal could provide a simple and general criterion for monitoring or automatically terminating training.

2 Related Topics

Our project builds upon several topics discussed during the course:

- Exploration through intrinsic motivation using Random Network Distillation (RND).
- Policy optimization using Proximal Policy Optimization (PPO) and value-based methods such as DQN.
- Measuring state-space coverage through state visitation entropy.

Relevant literature includes:

- [Burda et al., 2018]: *Exploration by Random Network Distillation*.
- [Schulman et al., 2017]: *Proximal Policy Optimization Algorithms*.
- [Bellemare et al., 2016]: *Unifying Count-Based Exploration and Intrinsic Motivation*.

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3 Idea

The main research question of this project is:

> Can the prediction error of Random Network Distillation serve as a reliable indicator of reinforcement learning convergence, and how does it compare to state visitation entropy?

Rather than using the RND prediction error as an intrinsic reward for exploration, we reinterpret it as a measure of state novelty throughout training. As the agent repeatedly encounters similar states, we hypothesize that the prediction error decreases and eventually stabilizes once learning has converged.

We will compare this signal with policy performance (episodic return) and state visitation entropy across multiple environments to evaluate whether RND prediction error consistently reflects learning progress.

Project for Reinforcement Learning lecture

Algorithm 1 Experimental procedure

Require: RL algorithm A, environment E, random seed s

Initialize policy and RND predictor

for each training iteration **do**

Collect experience in E

Update policy using A

Update RND predictor

Record:

- Episodic return
- Average RND prediction error
- State visitation entropy

end for

Compare convergence signals across multiple runs.

$$L_{\text{RND}} = \mathbb{E}_s \left[\|f(s) - \hat{f}(s)\|_2^2 \right] \tag{1}$$

4 Experiments

Our experiments aim to evaluate whether RND prediction error is a useful proxy for learning convergence.

Environments & Metrics We plan to evaluate our approach on two to three benchmark environments, such as Pendulum, MiniGrid, MarsRover, or BipedalWalker. During training, we will record episodic return, average RND prediction error, state visitation entropy, and the training timestep. These metrics will be compared to assess whether RND prediction error reliably reflects learning progress.

Experimental Scope For each environment, we will evaluate one to two reinforcement learning algorithms (e.g., PPO and DQN) using fixed hyperparameters and approximately five random seeds per setting. Additionally, we plan to evaluate a simple early stopping criterion based on the stabilization of the RND prediction error and compare it to a fixed training budget.

Estimated Computational Load The experiments will be conducted on relatively small benchmark environments that can be trained within a few hours. Since the project focuses on empirical analysis rather than large-scale performance, computational requirements remain moderate. Multiple random seeds can be executed in parallel.

5 Timeline

- Research and literature review: 3–4 days
- Experimental implementation: 1 week
- Running experiments: 1 week
- Analysis and visualization: 4–5 days
- Report writing and revisions: 1 week

References

Marc G. Bellemare, Sriram Srinivasan, Georg Ostrovski, Tom Schaul, David Saxton, and Remi Munos. Unifying count-based exploration and intrinsic motivation, 2016. URL `https://arxiv.org/abs/1606.01868`.

Yuri Burda, Harrison Edwards, Amos Storkey, and Oleg Klimov. Exploration by random network distillation, 2018. URL `https://arxiv.org/abs/1810.12894`.

John Schulman, Filip Wolski, Prafulla Dhariwal, Alec Radford, and Oleg Klimov. Proximal policy optimization algorithms, 2017. URL https://arxiv.org/abs/1707.06347.